

MLB ABS Challenge Run Value

Frozen 2026 regular-season snapshot through July 19

Research pipeline output

2026-07-20

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Provenance and validation

The analysis joins MLB live feeds, daily Statcast pitch exports, and Savant's ABS detail. Raw responses are cached with SHA-256 manifests. A challenge is published only when its identity and explicit outcome reconcile to both the game's final successful/failed totals and Savant.

Statcast alone is not the challenge ledger. Challenge timing and explicit overturned/upheld outcomes come from MLB's MJ review records; Statcast provides pitch geometry and the pre-pitch state. Initial and corrected post-call states are generated from that common state, and Savant independently validates the event identity, original call, outcome, and geometric ruling. The full acquisition and reconciliation history is recorded in [docs/data-acquisition-walkthrough.md](#).

Table 1: Publication acceptance checks

check	passed	observed	expected
challenge_resolution	TRUE	6263	6263
feed_total_reconciliation	TRUE	6263	6263
final_inventory_reconciliation	TRUE	2738	2738
savant_challenge_match	TRUE	6261	6261
feed_only_challenge_quarantine	TRUE	2	2
geometry_agreement	TRUE	6261	6261
manual_stratified_audit	TRUE	50	50

scope	team_id	candidates	covered	coverage
league	NA	219405	219171	0.9989

scheduled_games	observed_games	coverage
7289	7289	1

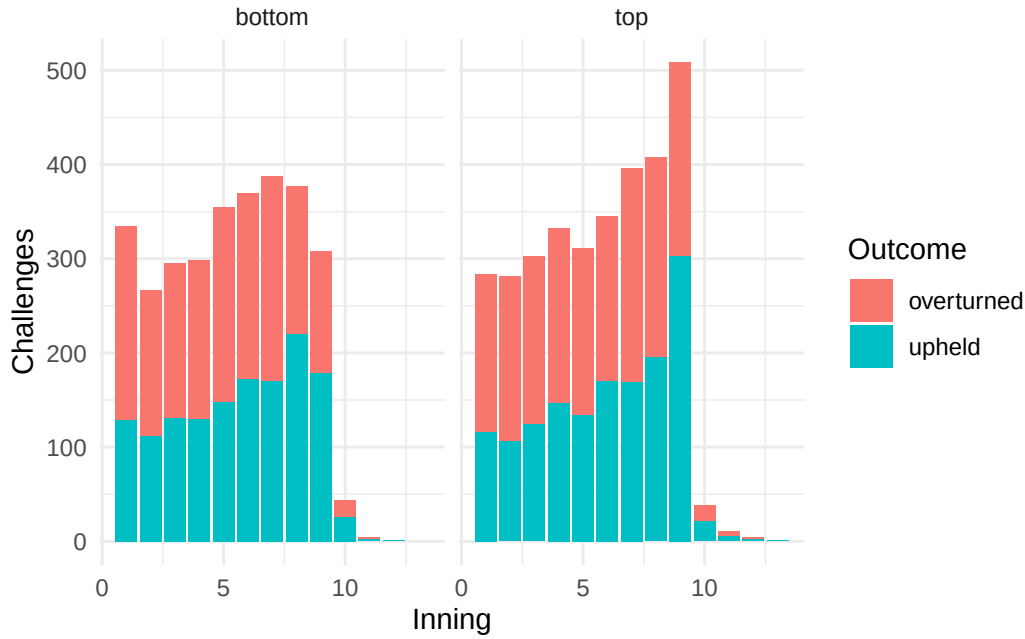
Table 4: Model validation diagnostics

diagnostic	n	metric_1	metric_2
RE288: 2025 holdout	712528	RMSE 0.9614	MAE 0.6235
Home WP: 2025 holdout	712528	Brier 0.1577	log loss 0.4700
RE288 vs Savant overturns	3343	RMSE 0.0492	correlation 0.9674

Challenge timing and success-rate accounting

Explicit **false** outcomes are upheld challenges, not missing observations. The denominator below is therefore every resolved attempt.

attempts	overturned	upheld	unresolved	success_rate
6261	3343	2918	0	0.5339



Actual challenge value

Immediate gain is the count-aware corrected-state minus original-state value from the challenger’s perspective. Upheld calls have zero immediate gain and their modeled marginal inventory cost is reported separately.

Table 6: League challenge value

at-tempts	over-turned	up-held	suc-cess_pct	gross_re	failed_inven-tory_re	net_re	gross_wpa	failed_inven-tory_wpa
6261	3343	2918	53.394	605.109	10.166	594.943	13.446	0.117

Table 7: Per-failure benchmark versus modeled inventory cost

upheld	Savant fixed RE	modeled inventory RE	modeled inventory WPA
2918	0.21	0.0035	0

Missed and out-of-challenges value

These are realized correctable calls. The out-of-challenges column does not assume that a club would have challenged every such pitch.

Table 8: Ten teams with the most realized RE lost at zero inventory

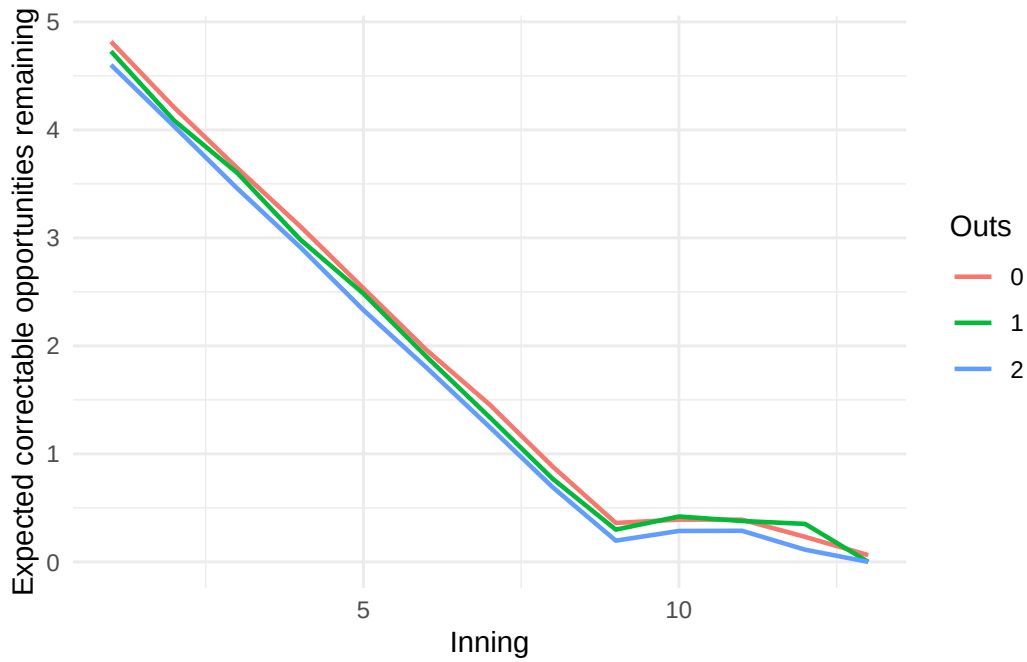
Team	Missed available	Missed RE	Out of challenges	Out RE
WSH	381	47.063	54	10.617
CLE	317	38.421	70	10.064
HOU	325	39.329	53	9.860
MIN	299	35.830	49	8.045
ATL	288	36.623	52	7.542
NYM	323	42.216	44	7.034
TB	338	44.202	43	7.026
MIL	396	47.716	44	6.826
LAA	380	49.457	49	6.667
KC	381	44.076	40	6.621

Remaining-opportunity inventory curves

Inventory is expressed first in future correctable opportunities, then RE and WPA. Curves use leave-one-game-out state averages; marginal values use the observed-behavior recursion.

Table 9: League-average remaining opportunity at game start

Correctable	Offense	Defense	Expected RE	Expected WPA	Unit 1 RE	Unit 2 RE
4.9517	2.5995	2.3522	0.6912	0.0101	0.01	0.0063



Team comparisons and WPA context

Raw totals and opportunity-adjusted attempt residuals are shown together; neither is interpreted causally.

Table 10: Challenge outcomes and net value by team

team	attempts	overturns	upheld	success_pct	net_re	net_wpa
CIN	228	148	80	64.912	29.971	0.595
MIN	283	161	122	56.890	29.966	0.729
COL	252	137	115	54.365	27.126	0.629
ATH	235	134	101	57.021	26.422	0.333
LAA	231	121	110	52.381	23.938	0.420
MIA	232	130	102	56.034	22.664	0.711
CWS	242	113	129	46.694	21.937	0.810
CHC	189	105	84	55.556	20.647	0.344
NYY	240	133	107	55.417	20.444	0.481
SEA	196	101	95	51.531	20.219	0.669
NYM	235	120	115	51.064	19.882	0.579
STL	182	100	82	54.945	19.734	0.859
AZ	172	109	63	63.372	19.712	0.402
BAL	195	98	97	50.256	19.626	0.473

team	attempts	overturns	upheld	success_pct	net_re	net_wpa
PHI	199	108	91	54.271	19.301	0.307
BOS	179	94	85	52.514	19.146	0.384
HOU	239	130	109	54.393	18.970	0.405
MIL	224	114	110	50.893	18.561	0.259
TEX	175	100	75	57.143	18.235	0.366
LAD	188	107	81	56.915	18.100	0.243
KC	205	115	90	56.098	17.881	0.429
ATL	191	97	94	50.785	17.693	0.337
DET	191	115	76	60.209	17.478	0.716
TOR	188	98	90	52.128	16.419	0.401
SF	188	91	97	48.404	16.326	0.289
SD	180	92	88	51.111	15.779	0.484
TB	199	103	96	51.759	15.683	0.056
CLE	212	93	119	43.868	14.943	0.049
WSH	196	93	103	47.449	14.865	0.170
PIT	195	83	112	42.564	13.275	0.402

Table 11: Missed, zero-inventory, and remaining-opportunity value

team	missed_re	out_re	expected_correctable	first_unit_re	second_unit_re
WSH	47.063	10.617	4.941	0.011	0.007
CLE	38.421	10.064	4.954	0.009	0.006
HOU	39.329	9.860	4.950	0.009	0.006
MIN	35.830	8.045	4.947	0.010	0.007
ATL	36.623	7.542	4.959	0.009	0.005
NYM	42.216	7.034	4.952	0.010	0.007
TB	44.202	7.026	4.947	0.010	0.006
MIL	47.716	6.826	4.952	0.011	0.007
LAA	49.457	6.667	4.951	0.012	0.008
KC	44.076	6.621	4.944	0.010	0.006
CWS	41.246	6.604	4.950	0.010	0.007
SF	34.557	6.142	4.963	0.009	0.005
PIT	45.593	5.762	4.950	0.009	0.006
NYN	32.505	5.724	4.961	0.008	0.005
ATH	47.975	5.592	4.947	0.011	0.007
COL	40.421	5.477	4.966	0.011	0.007
CHC	43.217	4.608	4.956	0.010	0.007
SD	44.032	4.177	4.949	0.010	0.006
MIA	42.664	4.085	4.946	0.010	0.006
DET	39.675	4.045	4.954	0.010	0.006

team	missed_re	out_re	expected_correctable	first_unit_re	second_unit_re
CIN	50.486	3.928	4.947	0.012	0.009
BAL	47.349	3.794	4.943	0.010	0.006
BOS	47.199	3.296	4.953	0.012	0.007
SEA	44.034	2.844	4.950	0.011	0.006
STL	45.593	2.410	4.950	0.011	0.006
TOR	32.232	2.310	4.948	0.007	0.005
PHI	41.777	1.785	4.953	0.009	0.006
AZ	38.588	1.650	4.953	0.008	0.005
TEX	47.634	1.330	4.961	0.011	0.006
LAD	47.860	1.246	4.953	0.009	0.006

Table 12: Opportunity-adjusted attempts with 95% game-cluster intervals

team	observed_attempts	expected_attempts	challenge_residual	lower	upper
MIN	283	201.272	81.728	50.211	120.886
COL	252	183.701	68.299	38.267	108.868
CWS	242	196.051	45.949	19.657	75.115
NYY	240	203.394	36.606	11.000	64.868
NYM	235	206.131	28.869	4.354	57.404
ATH	236	210.024	25.976	3.040	54.192
HOU	239	216.035	22.965	-6.838	54.288
MIA	232	212.604	19.396	-5.084	44.213
LAA	231	212.850	18.150	-6.012	42.580
CLE	212	203.577	8.423	-19.484	33.854
MIL	224	220.004	3.996	-22.693	30.842
TB	199	196.136	2.864	-19.671	25.445
ATL	191	189.983	1.017	-24.353	28.177
SEA	196	201.899	-5.899	-33.691	18.237
TOR	188	194.706	-6.706	-30.591	19.283
PIT	195	202.309	-7.309	-38.448	20.168
PHI	199	211.702	-12.702	-39.936	15.534
CIN	228	244.481	-16.481	-49.291	12.737
SF	188	205.179	-17.179	-43.119	9.618
BOS	179	196.466	-17.466	-42.252	8.923
LAD	188	208.348	-20.348	-42.292	1.673
KC	205	226.197	-21.197	-44.611	1.311
WSH	197	218.348	-21.348	-47.601	5.263
DET	191	212.369	-21.369	-48.915	5.438
SD	180	201.636	-21.636	-46.892	3.951
BAL	195	222.171	-27.171	-52.221	-1.502

team	observed_attempts	expected_attempts	challenge_residual	lower	upper
TEX	175	209.381	-34.381	-61.710	-7.605
CHC	189	223.945	-34.945	-62.936	-7.343
AZ	172	209.486	-37.486	-61.224	-11.377
STL	182	224.535	-42.535	-70.054	-14.839

Table 13: Net challenge RE with 95% game-cluster intervals

team	net_re	lower	upper
CIN	29.971	20.712	40.335
MIN	29.966	22.083	39.756
COL	27.126	18.770	36.169
ATH	26.422	19.232	33.374
LAA	23.938	15.787	32.895
MIA	22.664	16.077	29.499
CWS	21.937	13.783	31.162
CHC	20.647	13.556	28.886
NYN	20.444	14.307	26.332
SEA	20.219	13.059	28.685
NYM	19.882	14.019	26.243
STL	19.734	13.263	27.376
AZ	19.712	12.890	28.330
BAL	19.626	12.798	27.610
PHI	19.301	13.776	26.228
BOS	19.146	13.176	25.677
HOU	18.970	12.807	25.785
MIL	18.561	13.845	24.181
TEX	18.235	13.063	24.614
LAD	18.100	12.980	23.685
KC	17.881	12.955	23.346
ATL	17.693	11.901	24.919
DET	17.478	12.781	22.201
TOR	16.419	11.569	22.196
SF	16.326	11.149	22.685
SD	15.779	10.875	21.113
TB	15.683	11.275	20.575
CLE	14.943	10.369	20.268
WSH	14.865	10.007	19.086
PIT	13.275	9.084	18.331

Sensitivity and limitations

The boundary sensitivity removes pitches within 0.25 inches of the official ABS edge. This targets public-tracking and rounding uncertainty without reclassifying the remaining pitches.

correctable_all	correctable_outside_boundary	out_all	out_outside_boundary
14762	11931	1071	868

Unchallenged ABS outcomes remain geometry-based inferences. The RE288 and home-win GAM are empirical 2023–2025 models, and all adjusted comparisons may retain unmeasured differences in communication, pitcher/catcher confidence, and game context.